EXECUTIVE SUMMARY

Name of app: Malicious Mole Exposer

Roles of the team:

Clayton Pereira: Coding and Retraining of Neural Network

Husain Bin Mohamed Najid: Coding and Retraining of Neural Network

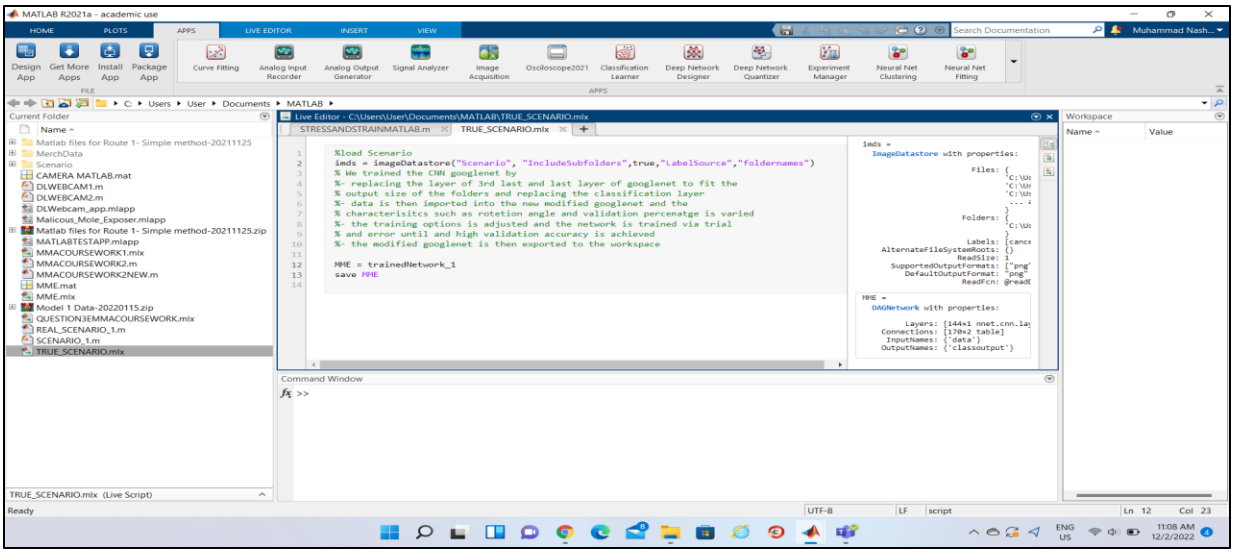
Hanfa Syed: Research and Presentation Preparation

User Persona

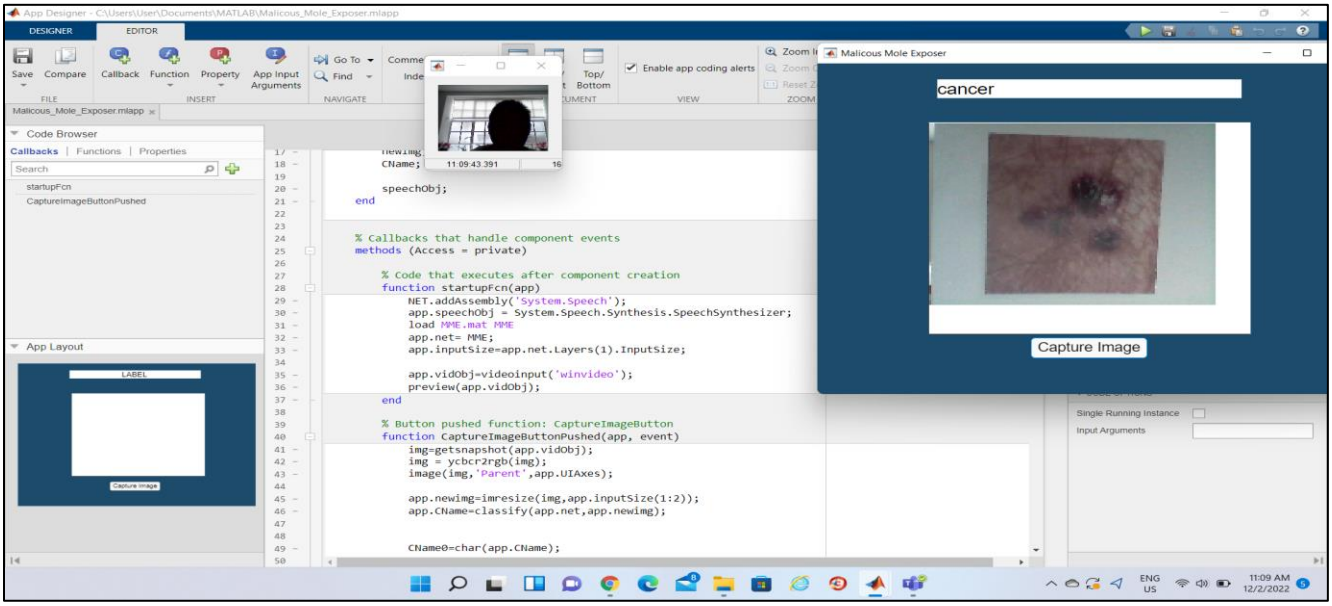
This app is targeted at all clients, specifically those who are more vulnerable to skin cancer. However, some people may be more vulnerable to cancerous moles than others, including people living in hotter countries like Australia and New Zealand as they're more exposed to ultraviolet radiation which increases chances of developing skin cancer. This is statistically supported as **" In 2007 in New Zealand, from a total population of 4 million people, 2,173 people were registered with melanoma and 292 people died from the disease [3**]. In **"Australia, a country of 21 million inhabitants, 10,342 people were diagnosed with melanoma and 1,279 died from it" [4].** It is also aimed at people who have many moles forming on the body. This app can therefore be used to classify and detect which moles are healthy and which could be potentially cancerous.

Coding and Functionalities of App

The images stored in the folder stored in the imageDatastore. The modification of the googlenet is not done by coding it but by configuring it in the Deep Network Designer in the App section of Matlab



After the images were saved in the datastore, we used the Deep Network Designer to remove the 3rd last layer's output to the number of files imported and the last classification layers. We then optimize and calibrate the training such as the number of images imported, the rotation angle of the pictures, the validation percentage, and the initial learn rate of the network. The network is then exported as a new trained network which named and saved as MME.mat. The MME.mat data is then loaded into the app as the new network and the app is ran.



Innovation Model

The application is a feasible idea since it can distinguish between a healthy and a cancerous mole, but more training is required of the neural network to increase its accuracy.

The product is desirable for oncologists in hospitals as it could provide faster diagnosis saving time and could eliminate the risk of human error. The app may be also desired by those who are at higher risk of developing skin cancers such as those who experience increased exposure to the sun and harmful UV radiation. They can use the app to quickly identify whether any mole they see on their bodies is malicious or not instead of visiting hospitals and wasting time if the mole turns out to be benign (non cancerous).

The app is also viable since it is has a potentially large market as anyone in the world can get cancer and skin cancer is a common form of cancer as well(1). The app itself is cheap to distribute since it costs $25 to publish an app of the Google Play Store while for Apple devices, it can cost $99 a year to publish.(2)

New Functionalities

One of the new functionalities learnt is the imageDatastore function which can import images in a folder and also all the image subfolders in we add the "IncludeSubfolders" function in the imageDatastore.

References

1. Cancer Research UK. *Melanoma skin cancer statistics*. https://www.cancerresearchuk.org/health-professional/cancer-statistics/statistics-by-cancer-type/melanoma-skin-cancer#heading-Zero. Accessed 12/02/2022
2. Appypie. *How much does it cost to publish an app on the app store?* https://www.appypie.com/faqs/how-much-does-it-cost-to-publish-an-app-on-the-app-store Accessed 12/02/2022
3. Sneyd M.J. & Cox, B. 2013. A comparison of trends in melanoma mortality in New Zealand and Australia: the two countries with the highest melanoma incidence and mortality in the world. *BMC Cancer, 13, 372.* https://doi.org/10.1186/1471-2407-13-372
4. Australian Institute of Health and Welfare. *Australian cancer incidence and mortality*. [http://www.aihw.gov.au/acim-books/] published on 06/08/2013